

ST JOHNS, AZ, USA

WMO: 723754

Lat: 34.518N

Lon: 109.379W

Elev: 5733

StdP: 11.90

Time Zone: -7.00 (W07)

Period: 99-19

WBAN: 93027

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.4	14.9	-7.4	4.6	36.2	-2.8	5.8	37.6	31.7	47.5	28.0	47.1	2.0	170	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	96.0	58.7	93.1	58.3	90.8	57.9	64.5	78.8	63.7	78.6	62.8	77.9	11.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	100.8	67.0	60.1	96.3	67.2	58.5	90.6	67.7	32.7	78.3	31.9	78.2	31.2	77.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.2	24.6	20.7	1.2	99.5	7.6	2.7	-4.2	101.4	-8.6	102.9	-12.9	104.4	-18.4	106.3		
			0.0	67.2	7.0	2.2	-5.1	68.8	-9.2	70.0	-13.2	71.3	-18.3	72.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	35.1	39.7	46.6	54.0	62.5	73.2	76.7	73.6	67.7	55.8	44.0	34.9	(6)
(7)		DBStd	16.03	7.71	7.37	7.11	6.42	7.07	4.75	3.82	3.57	4.72	6.67	7.46	7.63	(7)
(8)		HDD50	1638	463	293	147	35	5	0	0	0	25	202	468		(8)
(9)		HDD65	4550	927	709	570	332	129	4	0	1	25	291	630	932	(9)
(10)		CDD50	3610	1	3	42	155	393	695	827	733	532	205	22	2	(10)
(11)		CDD65	1046	0	0	0	2	52	249	362	269	106	6	0	0	(11)
(12)		CDH74	14064	0	0	33	172	1281	4069	3948	2741	1552	262	6	0	(12)
(13)		CDH80	6264	0	0	2	14	423	2104	1999	1177	509	36	0	0	(13)
(14)	Wind	WSAvg	7.5	5.8	7.6	8.6	10.3	9.6	9.0	7.5	6.4	6.5	6.5	6.2	6.0	(14)
(15)	Precipitation	PrecAvg	10.8	0.7	0.5	0.7	0.4	0.4	0.4	1.7	2.2	1.4	1.1	0.6	0.7	(15)
(16)		PrecMax	19.1	3.3	1.8	2.6	2.2	2.6	3.8	4.0	8.1	4.8	3.9	2.0	2.3	(16)
(17)		PrecMin	5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.0	0.6	0.5	0.6	0.5	0.5	0.7	1.1	1.4	1.1	0.9	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	70.0	78.3	82.1	92.6	98.3	100.4	94.9	90.8	84.0	74.3	64.1	(19)
(20)		0.4%	MCWB	43.7	45.2	49.2	50.7	57.3	57.7	61.0	60.3	58.5	55.1	48.1	42.9	(20)
(21)		2%	DB	59.1	65.6	73.3	78.8	88.2	95.5	97.0	91.9	88.2	80.6	70.5	59.3	(21)
(22)		2%	MCWB	41.2	43.6	46.6	49.0	54.4	57.0	59.8	59.8	57.6	53.6	46.8	41.8	(22)
(23)		5%	DB	54.9	61.2	69.9	75.4	84.4	93.1	93.4	89.8	85.7	77.0	66.2	54.7	(23)
(24)		5%	MCWB	39.8	41.5	45.0	47.6	52.2	56.4	59.8	59.8	57.1	51.7	45.5	39.9	(24)
(25)		10%	DB	51.1	56.5	65.6	72.8	81.5	90.8	90.5	87.1	82.8	73.3	62.2	50.7	(25)
(26)		10%	MCWB	38.2	39.8	43.6	46.7	51.7	55.8	59.8	60.0	56.4	50.2	44.0	38.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	47.8	50.7	52.3	58.5	62.0	66.3	66.0	63.2	59.0	51.2	46.7	(27)
(28)		0.4%	MCDWB	58.4	62.5	72.9	76.6	88.9	79.2	79.5	78.4	77.1	73.9	65.4	56.6	(28)
(29)		2%	WB	43.6	45.1	47.9	50.2	55.8	60.1	64.9	64.5	61.7	56.4	48.9	43.7	(29)
(30)		2%	MCDWB	54.5	61.1	69.8	74.9	82.6	83.9	79.2	78.5	75.8	72.0	64.3	55.1	(30)
(31)		5%	WB	41.2	43.1	46.1	48.7	54.0	58.9	64.0	63.8	60.6	54.3	46.9	41.3	(31)
(32)		5%	MCDWB	52.2	57.7	66.5	73.0	79.6	85.2	78.9	78.4	75.2	71.2	63.3	52.7	(32)
(33)		10%	WB	39.1	41.2	44.3	47.3	52.2	57.6	63.1	62.9	59.5	52.3	45.0	39.0	(33)
(34)		10%	MCDWB	49.2	54.8	63.1	70.5	77.9	85.1	78.5	77.6	74.7	69.8	60.9	49.6	(34)
(35)	Mean Daily Temperature Range		MDBR	28.1	28.9	31.9	32.2	32.2	33.8	27.1	26.4	29.3	31.3	31.5	27.2	(35)
(36)		5% DB	MCDBR	34.6	37.8	39.9	37.5	36.2	35.8	32.0	30.8	32.9	36.7	38.7	34.4	(36)
(37)		5% DB	MCWBR	20.3	20.2	19.6	17.5	15.4	13.5	9.4	8.7	11.2	16.4	20.4	20.2	(37)
(38)		5% WB	MCDBR	29.7	32.3	35.5	34.4	33.1	31.7	24.8	24.3	24.6	28.8	33.0	29.6	(38)
(39)		5% WB	MCWBR	18.1	17.4	17.6	16.2	14.5	12.0	6.8	7.2	8.6	13.1	18.0	18.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.245	0.269	0.283	0.296	0.306	0.359	0.346	0.307	0.273	0.247	0.238	(40)	
(41)		taud	2.649	2.642	2.548	2.495	2.503	2.472	2.432	2.432	2.533	2.599	2.636	2.652	(41)	
(42)		Ebn at Noon	318	325	321	318	313	308	291	294	303	309	311	312	(42)	
(43)		Edh at Noon	23	25	30	34	34	35	36	36	30	26	23	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1035	1336	1806	2243	2491	2575	2144	1994	1834	1519	1164	921	(44)	
(45)		RadStd	92	112	111	106	162	135	129	82	69	119	68	60	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-180	+131	+65

Nomenclature: See separate page